

Obsolete States Change Language-Model Decisions, and So Do Incidental Mentions

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Abstract

A model that reads an updated record can report the current value correctly and still let the old value shape its decision. We test whether this happens, and whether it depends on the old value having been superseded. In synthetic routing and access-control records, the current state, the policy and the correct action are fixed while one obsolete state is edited to imply the correct or the wrong action. Matched controls place the same word as an updated assignment of an unrelated attribute, as a word the entity merely mentioned, or as an unattached word. In a frozen confirmation on held-out wording (Qwen3-8B), we kept the routing histories in which the model retrieved the current and initial states and decided correctly without history (348 of 384). There, the obsolete state gave a net switch rate toward the wrong action of 0.158 (58 histories switched to wrong, 3 to correct). Across routing histories, a superseded assignment added 1.224 nats of wrong-action preference over an updated unrelated attribute. An incidental mention of the same word by the same entity moved decisions at least as much (net switch rate 0.278). A larger model from the same family, Qwen3-14B, replicated the obsolete-state effect and the supersession contrast; its mention effect matched the superseded one (net switch rates 0.169 and 0.167) rather than exceeding it. Gemma 3 4B did not meet the competence gate.

1 Introduction

Agents and assistants increasingly act on records that change: a user’s clearance is revoked, a ticket is rerouted, a preference is replaced. The record usually keeps the history, and the model is told that only the latest entry counts. Retrieval benchmarks ask whether the model can then report the current value, and recent work shows that old values do intrude on such reports, especially after many overwrites (Wang and Sun, 2026; Guo et al., 2026; Chat-taraj and Raj, 2026). A deployed system, however,

rarely asks the model to restate a value. It asks the model to *use* it. Two questions follow that retrieval accuracy does not answer. When a model can state the current value correctly, does the obsolete value still change the decision it takes? And if it does, is the obsolete value acting as a superseded assignment of the queried attribute, or simply as a word that occurs in the context near the entity?

The second question matters because the remedies differ. If the interference is tied to supersession, it is a failure of state tracking, and update-aware memory, historical marking or attention routing toward the current entry (Guo et al., 2026) are the natural fixes. If any nearby mention of a decision-relevant word does the same, the problem is contextual entrainment (Niu et al., 2025) reaching decisions. Removing or marking stale entries would then leave much of it in place, and evaluations that only vary update histories would misattribute it. The two accounts are not exclusive, and our design estimates both.

Design. We hold everything that should determine the decision fixed: the current state, the policy that maps states to actions, the query, and the correct action. We edit one obsolete state. In one member of each pair, the obsolete state implies the wrong action; in the other, it implies the correct action. The difference in the model’s preference for the wrong action between the two members is the decision effect of the obsolete content, and the paired generated answers give its behavioral counterpart (Figure 1). The same edit is repeated in matched constructions that keep the word and its position but change its role: an assignment to a different attribute that was later updated, a word the entity only mentioned, and a word attached to no entity. Before testing decisions, we check on the same records that the model reports the current and the initial state correctly when asked.

Findings. The experiment ran under a protocol frozen before the confirmation data were generated, on a held-out prompt template, with Qwen3-8B. A larger model from the same family, Qwen3-14B, was run on identical rows under a dated amendment, and Gemma 3 4B was attempted. The routing task meets every competence criterion on the confirmation data in both Qwen models; the access task does not, and its estimates are reported second and labeled. Three conclusions hold for both confirmed models, with one model-specific difference:

1. **Obsolete states change current decisions in models that track them correctly.** In the routing histories where the model retrieved the current and initial states and decided correctly without history, making the obsolete state imply the wrong action gave a net switch rate toward the wrong action of 0.158 (Qwen3-8B) and 0.165 (Qwen3-14B), almost never in reverse (§5.2, §5.5).
2. **Supersession adds decision pressure, and an incidental mention adds pressure of the same order.** In routing, a superseded assignment moved preference more than an updated assignment of an unrelated attribute, by 1.224 and 0.866 nats. The same word mentioned in passing by the same entity moved decisions as much as the superseded assignment in Qwen3-14B, and more in Qwen3-8B. Whether a mention *exceeds* a superseded assignment therefore depends on the model, and in Qwen3-8B also on wording (§5.3).
3. **A score-level sensitivity measure does not identify vulnerable items.** The edit’s effect on easy retrieval queries was large, but in both models it added nothing beyond the pre-decision features under the specified logistic model (§5.4).

What is and is not new. That stale context interferes with answers is established (Wang and Sun, 2026; Guo et al., 2026; Qian et al., 2026; Chao et al., 2026), as is the fact that mentioned tokens gain probability (Niu et al., 2025). Our contribution is a controlled test that connects the two at the level of decisions. We measure interference on policy decisions in open models, on items whose current and historical states the model retrieves correctly. We use minimal-pair edits that hold the

correct action fixed, report both scores and generated answers, and include the matched controls needed to ask whether supersession is the operative factor. We also report the negative result for a sensitivity-based predictor and, in Appendix A, two measurement problems we found along the way: chat-format events that the model never generates, and log-probabilities that change when tokens are appended under 8-bit and bfloat16 inference.

2 Related work

Interference from updated values. Proactive interference grows with the number of overwrites, and errors are mostly earlier values of the same key (Wang and Sun, 2026). Models retrieve the first value of a key more reliably than the latest (Chat-taraj and Raj, 2026; Qiao et al., 2026). Guo et al. (2026) name the failure *stale binding*. They find that the current value stays linearly recoverable on failures, that historical wording reduces errors, and that attention routing repairs most of them. Their decision scenarios, with closed reasoning models, showed no difference between full histories and resolved snapshots. Benchmarks of forgetting and stale memory evaluate dialogue and agent settings by accuracy (Qian et al., 2026; Chao et al., 2026). STALE separates recognizing an update from applying it. We use a single update, decisions fixed by an explicit policy, and matched minimal pairs with lexical controls, below the interference loads at which retrieval breaks.

Entrainment and distraction. Tokens that appear in the context gain probability whether or not they are relevant (Niu et al., 2025), and irrelevant context degrades reasoning (Shi et al., 2023). Our mention and unattached-word constructions measure this in the update setting. Counterfactual edits are compound treatments that need matched alternative edits as baselines (Yang et al., 2026b), which is how our contrasts are built.

Binding, state tracking and conflicts. Mechanistic accounts of entity binding (Feng and Steinhart, 2024; Gur-Arieh et al., 2026; Prakash et al., 2026) and state tracking (Tang et al., 2026; Kim and Schuster, 2023) may explain why lexical identity matters; we do not test them. Our records contain an intra-context conflict with an explicit temporal resolution (Xu et al., 2024). Questions about the value valid at a given time (Yang et al., 2026a) correspond to our historical retrieval check, not to our

Read the dated record. Only the final assignment governs the decision.
Initially, EvbUnit0096A had routing state pending. <- edited: pending | suspended
Initially, EvbUnit0096B had routing state suspended.
Later, EvbUnit0096A had archive tag restricted.
Finally, EvbUnit0096A had routing state restricted.
Later, EvbUnit0096B had archive tag restricted.
Finally, EvbUnit0096B had routing state pending.
Policy: state to bin: restricted -> B0, suspended -> B1, pending -> B2,
active -> B3; bin to action: B0 -> EAST, B1 -> EAST, B2 -> WEST, B3 -> WEST
Apply the supplied rule to the final assignment of EvbUnit0096A; give its action.
Respond with exactly one label (EAST, WEST), with no other text.

Construction	Edited line (same word, same position)
Superseded assignment	Initially, EvbUnit0096A had routing state pending.
Other attribute, updated	Initially, EvbUnit0096A had archive tag pending.
Mentioned by entity	Initially, EvbUnit0096A mentioned the word pending.
Unattached word	Initially, the unassigned word was pending.

Figure 1: One confirmation item (routing task, sequential order, held-out template; the annotation arrow is not part of the prompt). The current state restricted maps to EAST. The obsolete state is edited between pending (implies WEST, wrong) and suspended (implies EAST, correct); everything else is identical. The table shows how the edited line reads in each construction. The decision effect D_c compares the model’s preference for the wrong action across the two edits.

decision task.

3 Task, constructions and estimands

Records and decisions. Each history describes two entities. Each entity has an initial entry, an update of a separate *archive tag*, and a final assignment of the decision-relevant attribute (Figure 1). A policy maps four states to two actions, two states per action. In the *access* task, clearance maps directly to APPROVE or DENY. In the *routing* task, the routing state maps to a bin and the bin to EAST or WEST. The model is asked for the action implied by one entity’s final assignment and must answer with exactly one label. Two vocabularies are used: natural states (active, pending, restricted, suspended) and opaque states (S1–S4). Natural labels carry prior meanings that the policy may contradict. Three orders define the cells: *sequential* (both initial entries, then both final blocks), *interleaved* (initial and final blocks alternate across entities), and *competing* (interleaved, plus two archived discussions that quote decision-relevant state words for the two entities). The factorial crosses task, cell, vocabulary, correct action, historical order, current order and entity orientation: 192 histories per replicate. State-to-label assignments rotate so that every state appears in every role. In every history the other entity’s current state implies the wrong action. Confusing the two entities is therefore always an error, but it is identical in both members of a pair and cancels in the paired

contrasts below.

Constructions. For the queried entity, the obsolete line takes one of four forms, all with the same edited word in the same slot: *superseded* (the queried attribute was initially assigned the word and later reassigned); *updated other attribute* (the archive tag was initially the word and was later updated); *entity mention* (the entity mentioned the word, with no assignment); and *unattached* (the unassigned word was ...). Every history also contains a *current-only* item with no initial lines, which measures whether the model can apply the policy at all, and a *live* item that edits the final assignment itself, which shows the size of a legitimate change. The constructions share the policy, query, archive-tag update, final assignments and surrounding text. They differ in syntax around the edited word, so the contrasts compare constructions as wholes, not isolated semantic features.

Decision effect. For a prompt x , let $L(x) = \log P(a_{\text{wrong}} | x) - \log P(a_{\text{correct}} | x)$, where each action event is the label exactly as the model writes it at the start of its turn, followed by the end-of-turn token. For construction c and history h , the two members differ only in the obsolete word: in member $x_{h,c}^-$ it implies the wrong action, and in $x_{h,c}^+$ the correct one. The decision effect is

$$D_c = \frac{1}{|H|} \sum_{h \in H} [L(x_{h,c}^-) - L(x_{h,c}^+)]. \quad (1)$$

$D_c > 0$ means the obsolete content pushes preference toward the action it implies. It is a preference shift in nats, not an error rate. The two hypothesis contrasts are $\Delta_{\text{semantic}} = D_{\text{superseded}} - D_{\text{updated other}}$, which asks whether the assignment’s attribute matters, and $\Delta_{\text{mention}} = D_{\text{superseded}} - D_{\text{entity mention}}$, which asks whether assignment adds anything to an incidental mention.

Generated switches. With greedy decoding, a history scores +1 for construction c if the model answers wrongly under x^- and not under x^+ , -1 for the reverse, and 0 otherwise. The mean over histories is the *net switch rate*, a value in $[-1, 1]$. The counts of histories with +1 and -1 are reported as *histories switched* to the wrong and to the correct action.

Retrieval checks. On each record we also ask *easy* queries for the current state of the queried entity and of the other entity. For superseded items we ask for the *initial* state, under an introduction that announces both initial and final assignments. The answers are generated and parsed strictly.

Easy-query sensitivity. Let w^- and w^+ be the two values of the edited word, and let $m_x(w)$ be the log probability that easy query q , asked on record x , assigns to the answer w (summed over its spaced, unspaced and capitalized surface forms, scored as answer prefixes). Define $E(q) = [m_{x+}(w^+) - m_{x+}(w^-)] - [m_{x-}(w^+) - m_{x-}(w^-)]$. Then $R_{\text{easy}} = E(\text{queried entity}) - E(\text{other entity})$ is a per-history, query-specific sensitivity of retrieval scores to the edit. Hypothesis H3 asks whether R_{easy} helps predict downstream failures beyond features that are available before the decision query is asked.

4 Protocol

Model and runtime. Qwen3-8B (Yang et al., 2025) at a pinned revision (Appendix C), with thinking disabled through the chat template. Qwen3-14B and Gemma 3 4B were run later under dated amendments, with the same rows and rules (§5.5). Inference ran in float32 without quantization on one GPU. We chose float32 after finding that 8-bit and bfloat16 inference made the log probability at a fixed position change by up to 0.843 and 0.875 nats when tokens were appended after it. Float32 gave exactly zero (Appendix A.1). Action events were scored with a key-value-cache scorer. Before each run, the scorer was checked

against one full forward pass per event on identical prompts (maximum difference 5.08×10^{-5} nats against a tolerance of 0.02). Answers were generated greedily with at most 16 new tokens and parsed strictly: an answer counts only if, after stripping outer whitespace, it is exactly one allowed label.

Splits and gate. Three disjoint splits were generated from separate seeds and separate entity-name pools. A development split (192 histories) was used only to check competence and answer formats. A frozen gate split (384 histories) decided eligibility. The confirmation split (768 histories, 8,448 records, 11 per history) used a held-out paraphrase of the introduction and query. A task-cell was *competent* if strict direct retrieval of the current state was at least 0.95, current-only decision accuracy at least 0.90, and invalid answers at most 0.05. Confirmation data were generated only after the gate, recomputed from its saved scores, showed at least one competent cell. The gate decided eligibility; competence labels in the results are computed on the confirmation data, except in Table 7, which reports the gate. All cells are reported. Task-level estimates average the fixed grid, and estimates from non-competent cells are labeled as such.

Hypotheses and statistics. The protocol fixed two primary hypotheses per task: H1, $D_{\text{superseded}} > 0$; and H2, $\Delta_{\text{semantic}} \neq 0$, two-sided. Both use 97.5% percentile intervals from a history bootstrap (2,000 draws, fixed seed), Bonferroni-adjusted over the two tasks. Secondary outcomes use unadjusted 95% intervals: Δ_{mention} , the other construction effects, net switch rates, and H3. Histories are the resampling unit. Both members, all constructions and all queries of a history stay together. The two tasks are never pooled, and an interval that includes zero is reported as inconclusive, not as evidence of no effect.

Prediction (H3). Predictors that are available before the decision query is asked: the easy query’s margin between the current and the historical state, the historical candidate’s log mass, easy confidence, the edited token’s position, prompt length, and design indicators. The comparison adds R_{easy} to these. The outcome is a strict downstream failure on the historical constructions. Models are L2-regularized logistic regressions (scikit-learn; Pedregosa et al., 2011) ($C = 1$, no tuning) evaluated

Cell	Direct	No history	History	Competent
routing seq.	0.988	0.953	0.858	yes
routing int.	0.986	0.961	0.868	yes
routing comp.	0.982	0.953	0.846	yes
access seq.	0.916	0.867	0.831	no
access int.	0.945	0.875	0.858	no
access comp.	0.908	0.859	0.867	no

Table 1: Strict accuracy on the confirmation data. *Direct*: current state of the queried entity, asked directly. *No history*: decision on the current-only item. *History*: decision over all historical-construction items. *Competent*: direct ≥ 0.95 , no history ≥ 0.90 , invalid ≤ 0.05 . No answer was invalid.

by five-fold cross-validation with folds grouped by history. Out-of-fold log loss with and without R_{easy} is compared by a paired history bootstrap. The downstream answer’s own confidence is reported separately as a contemporaneous reference, not as a predictor, because it comes from the decision query itself.

Protocol history. The confirmation protocol is the second redesign of a design frozen before any model was run; every change, and the amendments that added models, are dated and listed in Appendix A. The records are internal, so we call them *frozen before confirmation*, not preregistered.

5 Results

All numbers in this section come from the frozen confirmation (Qwen3-8B, held-out template, 384 histories per task) unless marked otherwise.

5.1 Competence

On the confirmation data, the routing cells meet all competence criteria and the access cells do not, mainly because current-only decisions fall short of 0.90 (Table 1). Asked for the *initial* state on the superseded items, the model answered correctly in 0.990 of routing and 0.973 of access cases. In routing, then, the model reports the current state, reports the initial state, and applies the policy without history at or above 0.95. Access results are reported below as fixed-grid averages over cells that do not meet the competence criteria.

5.2 Obsolete states change decisions

H1 is supported in both tasks. In routing, when the superseded state implied the wrong action rather than the correct one, the wrong action’s log-odds

rose by $D_{\text{superseded}} = 3.133$ nats (97.5% interval [2.743, 3.516]). In access, where cells do not meet the competence criteria, the shift was 3.904 [3.474, 4.346]. For scale, editing the *current* state itself, which legitimately changes the correct action, moved the same quantity by -8.197 (routing) and -10.100 nats (access).

The shift reaches generated answers. In routing, the net switch rate toward the wrong action was 0.185 [0.143, 0.227]: 75 histories switched to the wrong action and 4 to the correct one. In each routing cell, it was between 0.164 and 0.227 (Table 4). In access it was 0.167 [0.130, 0.206] (65 against 1). Figure 3 shows the same effect as accuracy. In every routing cell, decisions were correct in 0.906–0.953 of histories when the superseded state implied the correct action, and in 0.727–0.750 when it implied the wrong one. The same records gave current- and initial-state retrieval at or above 0.98.

Histories the model demonstrably tracks. Aggregate retrieval accuracy does not show that the switching histories are ones the model reads correctly. We therefore kept a history only if three conditions held on the same records: the model retrieved the current state correctly under both members of the pair, retrieved the initial state correctly under both superseded members, and decided the current-only item correctly. This kept 348 of 384 routing histories for the superseded construction (Table 2). In them, the net switch rate toward the wrong action was 0.158 [0.118, 0.198]: 58 histories switched to the wrong action and 3 to the correct one. The preference shift was 2.990 nats [2.661, 3.342]. Each routing cell keeps a net switch rate between 0.129 and 0.202. The effect is therefore not carried by histories in which the model misreads the record. In access, the same restriction keeps 284 of 384 histories and gives a net switch rate of 0.176 [0.134, 0.218] (50 against 0).

5.3 Supersession adds pressure; a mention adds comparable pressure

This section reports Qwen3-8B, where a mention adds more pressure than a superseded assignment; §5.5 shows that in Qwen3-14B it adds comparable pressure. H2 compares the superseded assignment with an assignment of the same word to a different attribute that was later updated. It is supported in routing, $\Delta_{\text{semantic}} = +1.224$ [0.973, 1.485], with generated switches agreeing

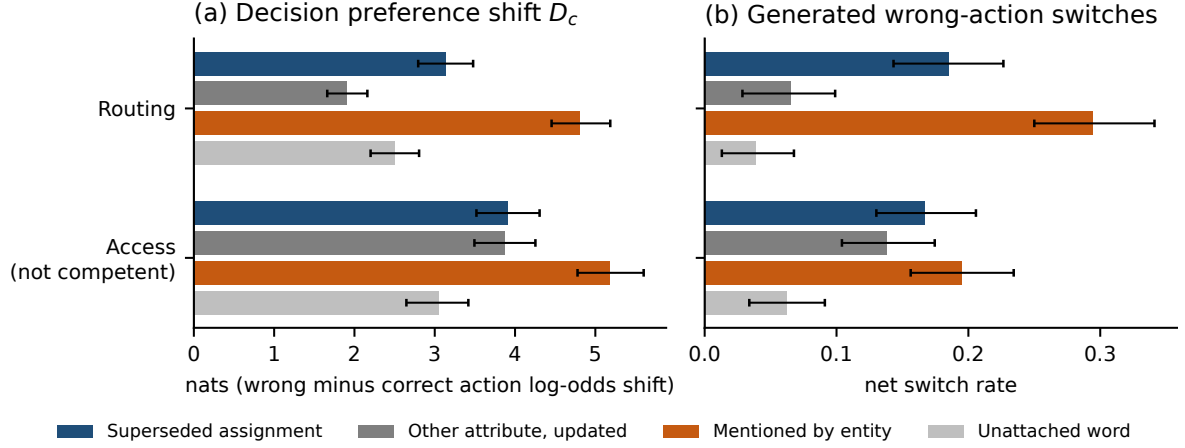


Figure 2: Decision effects by construction (confirmation, 384 histories per task, 95% history-bootstrap intervals). (a) D_c : increase in the wrong action’s log-odds when the edited word implies the wrong rather than the correct action. (b) Net switch rate toward the wrong action (mean over histories of +1, 0 or −1). Routing cells meet the competence criteria on the confirmation data; access cells do not.

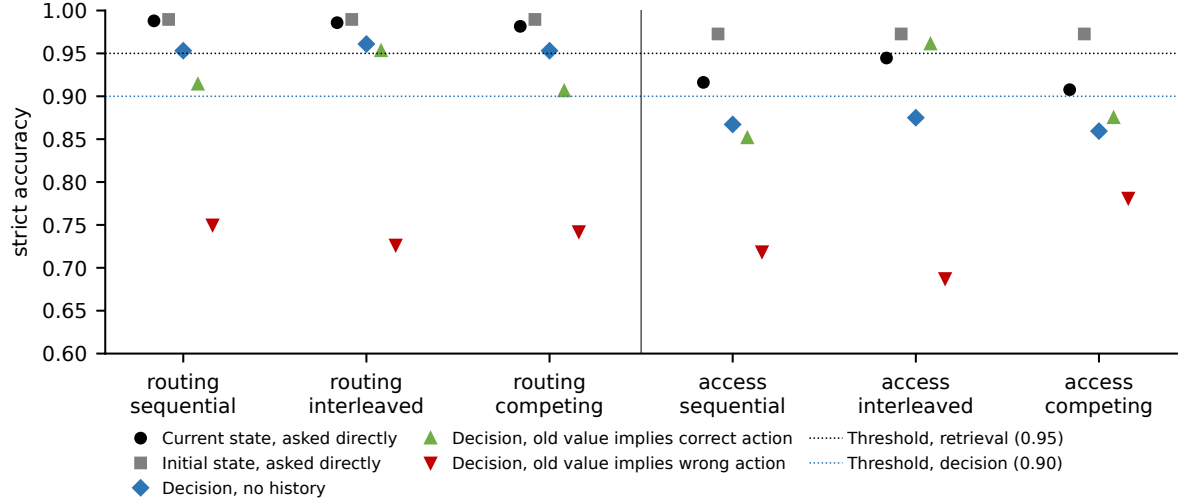


Figure 3: Retrieval and decision accuracy on the same confirmation records, per cell. Initial-state retrieval is reported per task. Routing (left) meets the competence criteria on the confirmation data; access (right) does not.

(+0.120 [0.076, 0.164]). In routing, then, the attribute that was superseded matters. In access (non-competent cells) the contrast is inconclusive: +0.035 [−0.318, 0.400], switches +0.029 [−0.016, 0.073].

The mention control, a secondary outcome, shows that supersession is not the largest source of pressure in Qwen3-8B. When the same entity only *mentioned* the word, the effect was larger than for the superseded assignment. In routing, $D_{\text{entity mention}} = 4.807$ against 3.133, so $\Delta_{\text{mention}} = -1.675$ [−1.896, −1.441]. Net switches were 0.294 (113 to wrong, 0 to correct) against 0.185, difference −0.109

[−0.156, −0.065]. In access, 5.182 against 3.904 ($\Delta_{\text{mention}} = -1.278$ [−1.579, −0.977]). Switches there were 0.195 (76 against 1) against 0.167, an inconclusive difference. A word attached to no entity still shifted preference substantially, by 2.496 (routing) and 3.043 nats (access), but rarely changed a generated decision (0.039 and 0.062). Score shifts of similar size can thus have very different behavioral consequences. The ordering holds in the restricted routing histories of Table 2. Net switch rates there are 0.278 for the mention, 0.158 for the superseded assignment, 0.048 for the updated unrelated attribute and 0.034 for the unattached word.

Across four earlier, non-confirmatory splits,

	Kept	$D_{\text{superseded}}$	Net switch rate	Switched
routing seq.	118/128	2.959	0.144 [0.076, 0.220]	19 / 2
routing int.	114/128	3.252	0.202 [0.132, 0.281]	24 / 1
routing comp.	116/128	2.764	0.129 [0.069, 0.190]	15 / 0
routing, all	348/384	2.990	0.158 [0.118, 0.198]	58 / 3
access, all	284/384	4.059	0.176 [0.134, 0.218]	50 / 0

Table 2: Superseded construction, restricted to histories with correct current-state retrieval under both members, correct initial-state retrieval under both members, and a correct current-only decision. Kept: histories retained out of those in the cell. Switched: histories switched to the wrong / to the correct action. 95% history-bootstrap intervals. Retained sets differ slightly across constructions because condition (i) is checked per construction. Condition (ii) is checked on the history’s superseded members for every construction, because the initial-state query is asked only there.

Δ_{mention} never favored the superseded assignment, but its size varied with the template (Appendix B, Figure 4).

5.4 Easy-query sensitivity adds nothing to failure prediction

The edit strongly affected easy retrieval scores: the mean R_{easy} on superseded items was 8.096 nats in access and 10.878 in routing. That sensitivity added nothing beyond the pre-decision features under the specified logistic model (Table 6 in Appendix B). Pre-decision features reduced log loss from 0.414 to 0.346 and reached AUROC 0.780. Adding R_{easy} changed log loss by +0.0004 nats, interval $[-0.0011, 0.0019]$; the upper end is under 3% of what the simple features already gain. The result is the same in each task and for the wrong-answer outcome. It replicates in Qwen3-14B: AUROC 0.684 without and 0.685 with R_{easy} , log-loss gain $-0.0001 [-0.0027, 0.0022]$. Within this design, R_{easy} describes how retrieval scores respond to the edit; it does not diagnose decision vulnerability.

5.5 Replication in Qwen3-14B, and a model that was not eligible

Qwen3-14B was added under an amendment dated before any of its data were scored, with the same rows, rules and runtime (Appendix A). Its frozen gate passed in all three routing cells and in no access cell (Table 7). On the confirmation, routing retrieval was 0.989–0.997 for the current state and 0.997 for the initial state. Current-only decisions were 0.977–0.992 correct.

Table 3 compares the two models on the cells

competent on the confirmation data. The obsolete-state effect replicates. In routing, $D_{\text{superseded}} = 4.785$ nats and the net switch rate is 0.167 (67 histories switched to wrong, 3 to correct). In the routing histories Qwen3-14B demonstrably tracks (376 of 384), it is 0.165 [0.128, 0.205] (Table 8). Supersession again adds preference over an updated unrelated attribute: $\Delta_{\text{semantic}} = +0.866 [0.417, 1.328]$ in routing. In generated switches, however, the two constructions are close in this model (0.167 against 0.146). The mention result differs from Qwen3-8B. In Qwen3-14B the mention moves routing decisions as much as the superseded assignment does: net switch rates are 0.169 against 0.167 over all routing histories, and 0.168 against 0.165 in the tracked histories (Table 8), with $\Delta_{\text{mention}} = +0.248 [-0.117, 0.601]$, an inconclusive difference. In access (non-competent cells) the superseded assignment moved preference more than the mention ($\Delta_{\text{mention}} = +0.845 [0.425, 1.255]$). In both models, then, an incidental mention produces interference of the same order as a superseded assignment. Whether it exceeds it is model-specific.

Gemma 3 4B was added under a model list frozen before any of its data were scored. On the gate split, no cell met the criteria: current-only decision accuracy was 0.609–0.719, direct retrieval 0.916–0.964 (Table 7), while initial-state retrieval in development was 0.984. Gemma reads these records but does not apply the policy reliably, so no confirmation split was generated for it. This is an eligibility result, not evidence for or against stale-state interference in Gemma.

6 Discussion

What the evidence supports. In two models that report both the current and the initial state correctly, an obsolete state still moves routing decisions toward the action it implies. The net switch rate is 0.158 (Qwen3-8B) and 0.165 (Qwen3-14B) in the histories where the model retrieved both states and decided correctly without history. Stale-value interference, documented for value queries (Guo et al., 2026; Wang and Sun, 2026), thus also reaches single-step policy decisions in open models after a single update, on items where retrieval is at ceiling. Retrieval accuracy would not reveal this gap. The controls separate two sources of the effect. Supersession contributes: in routing, a superseded assignment of the queried attribute moved preference

Model	Task (competent cells)	n	$D_{\text{superseded}}$	Δ_{semantic}	Δ_{mention}	Net switch rate
Qwen3-8B	routing (comp., int., seq.)	384	+3.133 [+2.743, +3.516]	+1.224 [+0.973, +1.485]	-1.675 [-1.896, -1.441]	+0.185 [+0.143, +0.227]
Qwen3-8B	access	no competent cell				
Qwen3-14B	routing (comp., int., seq.)	384	+4.785 [+4.244, +5.354]	+0.866 [+0.417, +1.328]	+0.248 [-0.117, +0.601]	+0.167 [+0.130, +0.206]
Qwen3-14B	access	no competent cell				
Gemma 3 4B	not eligible: no competent cell at the gate					

Table 3: Confirmation results per model, on the task-cells that met the competence criteria *on the confirmation data* (the rule used in the text and in Table 1). D and Δ in nats; $D_{\text{superseded}}$ and Δ_{semantic} with 97.5% intervals, Δ_{mention} and the superseded net switch rate with 95% intervals (history bootstrap). Models are reported separately, never pooled. The replication amendment defined competent cells by each model’s frozen gate (Table 7); gate-rule estimates are in Appendix B.

more than an updated assignment of an unrelated attribute in both models (1.224 and 0.866 nats). But the same word, mentioned once by the same entity and never assigned, moves decisions about as much in Qwen3-14B and more in Qwen3-8B. The interference therefore does not require supersession. It behaves like entrainment by decision-relevant words attached to the queried entity (Niu et al., 2025), with supersession adding to it. How the two compare is model-specific.

Consequences for evaluation and mitigation.

Two practical points follow. First, stale-state benchmarks that vary only update histories cannot tell supersession from incidental mention. A matched mention condition is cheap and changed our conclusion. Second, mitigations should be evaluated the same way: with a matched mention control and decision-level outcomes, not retrieval accuracy alone. We did not test a mitigation. An evaluation should also check historical retrieval, which the model performs well and which a blunt fix could damage.

Scores and behavior. Two score-level results caution against reading log-probability shifts as behavior. An unattached word shifted preferences by 2.5–3.0 nats but changed few decisions (net switch rates 0.039 and 0.062). And retrieval-score sensitivity to the edit, although large, added nothing beyond the pre-decision features in predicting which decisions failed. A preference shift matters behaviorally only when it crosses the item’s decision margin, so it can be large and still inconsequential. Studies that measure only scores, or only accuracy, see different parts of the same effect.

Alternative explanations. The constructions differ in syntax around the edited word, so the contrasts compare whole constructions. Correct direct retrieval also does not show that the current state

was the one used at decision time, so the retrieval checks do not rule out stale binding in the sense of Guo et al. (2026). The mention construction could be read as more salient. Its advantage varied with template in Qwen3-8B and did not appear in Qwen3-14B, which fits a surface explanation better than a categorical one. The other entity’s current state always implied the wrong action. Confusing the two entities therefore always yields the wrong action. This may contribute to baseline errors, but it cancels in every paired contrast. Natural state words carry prior meanings, but the pattern does not depend on them. For Qwen3-8B, in all 12 task $\times$ cell $\times$ vocabulary strata (64 histories each), $D_{\text{superseded}}$ was positive (2.53–4.53 nats). Δ_{mention} was negative in all 12, with intervals excluding zero in 11, including all six with opaque labels (Table 5).

7 Conclusion

In two open models that report the current and the initial state of a record correctly, an obsolete state still pulls routing decisions toward the action it implies (net switch rates 0.158 and 0.165 in histories the models demonstrably track). Supersession adds to this pull, but is not required: the same word mentioned in passing by the same entity pushes about as hard, harder in one model. A score-level measure of retrieval sensitivity added nothing to failure prediction beyond pre-decision features under the specified logistic model. Stale-context benchmarks and mitigations should therefore be evaluated with incidental-mention controls and decision-level outcomes, not only retrieval accuracy. Both confirmed models come from one family. Gemma 3 4B could not perform the base task reliably enough to be evaluated, and other families remain untested.

Limitations

Two models, one family. The confirmed results are for Qwen3-8B and Qwen3-14B. Gemma 3 4B did not meet the competence gate, and four further models listed in our amendment were deferred. We make no claim about other model families, and the mention comparison already differs between the two Qwen models.

Synthetic tasks. The records, policies and constructions are synthetic and short: two entities, one update and an explicit policy. This control is what lets the contrasts hold the correct action fixed, but the effect sizes need not transfer to natural dialogue, long agent logs or tool outputs.

Competence in access. The access cells did not meet the competence criteria on the confirmation data in either model, mainly because current-only accuracy was 0.859–0.875 for Qwen3-8B and 0.820–0.859 for Qwen3-14B. Access estimates are fixed-grid averages over cells where the model already makes errors without history.

Construction differences. Constructions share the edited word and its position but differ in syntax. The contrasts compare constructions, not isolated semantic properties. The template dependence of the mention contrast shows that wording matters.

Behavioral evidence only. We measure scores and generated answers. Nothing here identifies a mechanism, and terms such as *entrainment* describe the behavioral pattern, not a verified internal process.

Prediction scope. H3 uses cross-validation within the confirmation split. A null result there does not rule out other score-based diagnostics, other outcomes, or other models.

No mitigation. We did not test an intervention; the paper makes no claim about which mitigations work.

Use of AI assistants. A large language model assisted with implementing the experiment code and drafting this manuscript. The authors checked the claims, analyses, references and final text.

Ethical Considerations

All data are synthetic records generated from templates. No human subjects, annotators or personal data were involved. The results bear on systems

that act on changing records, such as access control and routing. They show that an incidental mention near an entity can shift an automated decision. This could be used deliberately: someone able to insert text into a record could bias a model-mediated decision without changing any authoritative value. We report the effect so that evaluations and safeguards can account for it. Our experiments do not test any deployed system. The models were used for research under their published licenses (Appendix C).

References

- H. Chao, Y. Bai, R. Sheng, T. Li, and Y. Sun. 2026. [STALE: Can LLM agents know when their memories are no longer valid?](#) *arXiv preprint arXiv:2605.06527*.
- S. Chattaraj and K. Raj. 2026. [LLMs remember first, forget last: Dual-process interference in large language models](#). *arXiv preprint arXiv:2603.00270*.
- T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer. 2022. [LLM.int8\(\): 8-bit matrix multiplication for transformers at scale](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- J. Feng and J. Steinhardt. 2024. [How do language models bind entities in context?](#) In *International Conference on Learning Representations (ICLR)*, pages 36391–36413.
- J. Guo, Y. Fang, S. Gu, C. Spanos, J. Demmel, and J. Lavaei. 2026. [When context changes: Understanding update failures in LLMs](#). *arXiv preprint arXiv:2609.38866*.
- Y. Gur-Arieh, M. Geva, and A. Geiger. 2026. [Mixing mechanisms: How language models retrieve bound entities in-context](#). In *International Conference on Learning Representations (ICLR)*, pages 27757–27786.
- N. Kim and S. Schuster. 2023. [Entity tracking in language models](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3835–3855.
- J. Niu, X. Yuan, T. Wang, H. Saghir, and A. H. Abdi. 2025. [Llama see, llama do: A mechanistic perspective on contextual entrainment and distraction in LLMs](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16218–16239.
- A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, and 2 others. 2019. [PyTorch: An](#)

imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32.

F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

N. Prakash, N. Shapira, A. Sen Sharma, C. Riedl, Y. Belinkov, T. Rott Shaham, D. Bau, and A. Geiger. 2026. Language models use lookbacks to track beliefs. In *International Conference on Learning Representations (ICLR)*, pages 7657–7694.

Y. Qian, Z. Yang, W. Ding, H. Li, and Y. Xie. 2026. Do LLMs forget what they should? evaluating in-context forgetting in large language models. In *International Conference on Learning Representations (ICLR)*, pages 108956–108983.

B. Qiao, S. Guo, X. Yang, K. Li, W. Zhou, S. Hu, and Y. Song. 2026. Diagnosing retrieval bias under multiple in-context knowledge updates in large language models. *arXiv preprint arXiv:2603.12271*.

F. Shi, X. Chen, K. Misra, N. Scales, D. Dohan, E. H. Chi, N. Schärli, and D. Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 31210–31227.

Z. Tang, Q. Zhao, G. Franco, D. T. Wijaya, A. Mueller, S. Schuster, and N. Kim. 2026. Do language models track entities across state changes? In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 119466–119503.

C. Wang and J. V. Sun. 2026. Unable to forget: Proactive interference reveals working memory limits in LLMs beyond context length. In *Conference on Language Modeling (COLM)*.

T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, and 3 others. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45.

R. Xu, Z. Qi, Z. Guo, C. Wang, H. Wang, Y. Zhang, and W. Xu. 2024. Knowledge conflicts for LLMs: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8541–8565.

A. Yang et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

X. Yang, Z. Liu, R. Li, and Y. Lei. 2026a. Large language models in resolving contextual knowledge conflicts. *arXiv preprint arXiv:2609.03148*.

Z. Yang, M. Levy, Y. Goldberg, and B. C. Wallace. 2026b. Compared to what? baselines and metrics for counterfactual prompting. In *Conference on Language Modeling (COLM)*.

A Protocol history

Every change below is recorded, with its date, in the protocol documents released with the code. None was made after looking at confirmation data.

v1 (frozen design, never run). The design fixed the hypotheses, estimands, factorial, constructions, gates, bootstrap and held-out confirmation template. An audit before any inference found two problems. (i) The action events began with a space (`_APPROVE`). Qwen3-8B begins its turn without one, and the two strings tokenize differently, so v1 would have scored continuations the model never produces. (ii) Among the H3 predictors was the decision’s own action confidence, which is not available before the decision is made.

v1b (corrected before inference). Action events became the label exactly as generated plus the end-of-turn token. The H3 predictors were split into pre-decision predictors and a contemporaneous reference. Historical retrieval was added as a competence measure. During setup, the runtime was changed twice for numerical reasons (Appendix A.1), and a rounding tolerance of 10^{-5} was added to validators that rejected float32 log probabilities a few 10^{-7} above zero. The v1b gate failed: direct retrieval was 0.851–0.946 in every cell. The cause was the prompt format. The user message ended with a literal `Answer: line`, which the model often echoed (about 260 of 4,224 retrieval answers); about 100 answers gave an action label instead of a state. Only 64 named the obsolete value. The decision effects replicated on the gate histories.

v2 (first redesign; at most two were allowed). The redesign changed only what the failure required: it removed the trailing `Answer: line` and used new seeds and entity-name pools so that all histories were new. Confirmation size was set to four replicates (768 histories) before any v2 data existed, to fit the compute budget. The scorer was

changed from a batched scorer to a key-value-cache scorer for speed, before any v2 effect was inspected. Both scorers compute the same quantity, and the cache scorer passed its equivalence check. The v2 gate passed in three cells, and confirmation data were then generated.

Second model. The model list, adding Gemma 3 4B with the same rows, rules and runtime, was fixed before any Gemma data were scored. A 24-record smoke test, inspected for format only, showed that Gemma writes its label, a newline and then its end-of-turn token. The scored events had omitted the newline and covered at most a few percent of the probability mass. The scored event was corrected to include the newline, the checks were rerun, and the smoke outputs were set aside. Development and gate then failed the competence criteria in every cell (§5.5).

Replication models. A further amendment, dated before any of their data were generated, listed five models in a fixed order with pinned revisions. It fixed the per-model sequence, the competence criteria, the reporting rule, and the definition of competent cells: those passing on the model’s own frozen gate. For this submission the scope was reduced, before any of the five had produced data, to the first model, Qwen3-14B. The other four are deferred under the same amendment. Qwen3-14B’s preflight needed no correction (causality 0.0; scorer difference 8.3×10^{-7} nats; it ends its answer with the label and its end-of-turn token). Its rows were byte-identical to Qwen3-8B’s in every split.

A.1 Numerical causality of log-probability scoring

Scoring answers by teacher forcing assumes that the log probability at a position does not depend on tokens appended after it. We tested this directly on the same prompts, by comparing the next-token distribution at a fixed position with and without one or two tokens appended. With 8-bit weights (LLM.int8(), Dettmers et al., 2022), log probabilities changed by up to 0.843 nats. With bfloat16, the top-20 log probabilities changed by up to 0.875 nats, and computing only the output layer in float32 reduced it to 0.502. In float32, and with 4-bit nf4 weights, it was exactly zero. In practice this broke our validators: four disjoint candidate answers summed to more than probability one. A key-value-cache scorer also disagreed with full re-computation by 3.406 nats under 8-bit weights, but

by 5.08×10^{-5} in float32. Score differences of a few nats are the size of the effects studied here, so we ran all inference in float32. Score-based studies that use reduced precision should report a check of this kind.

B Additional results

This appendix gives the per-cell switch rates behind §5.2 (Table 4), the H3 prediction table (Table 6), and the per-stratum decision effects referred to in §6 (Table 5). It also shows the replication of the primary and mention contrasts across all splits (Figure 4, summarized in §5.3), the gate results of every model attempted (Table 7), the tracked-history analysis per model (Table 8), and Qwen3-14B’s construction effects (Figure 5). Rows are ordered with routing, the task competent on the confirmation data, first.

Cell	Superseded	Other attr.	Mention	Unattached
routing seq.	0.164	0.062	0.352	0.055
routing int.	0.227	0.078	0.273	0.055
routing comp.	0.164	0.055	0.258	0.008
access seq.	0.133	0.070	0.148	0.094
access int.	0.273	0.234	0.219	0.094
access comp.	0.094	0.109	0.219	0.000

Table 4: Net switch rate toward the wrong action, per cell and construction (confirmation, $n = 128$ histories per cell). Intervals are in the released analysis.

Stratum	Vocab.	$D_{\text{superseded}}$	Δ_{mention}
routing seq.	natural	3.66	−2.32
routing int.	natural	4.03	−1.00
routing comp.	natural	3.07	−2.38
routing seq.	opaque	2.53	−2.17
routing int.	opaque	2.78	−1.32
routing comp.	opaque	2.73	−0.87
access seq.	natural	3.61	−1.16
access int.	natural	4.01	−0.06
access comp.	natural	3.65	−2.21
access seq.	opaque	4.53	−1.10
access int.	opaque	4.43	−1.42
access comp.	opaque	3.20	−1.72

Table 5: Decision effects (nats) by task, cell and vocabulary (confirmation, 64 histories per stratum). All $D_{\text{superseded}}$ intervals exclude zero. The Δ_{mention} interval includes zero only for access, interleaved, natural ([−1.01, 0.96]).

Replication across splits. Four earlier, non-confirmatory splits of the same protocol family (development and gate of the first redesign and of the confirmed protocol, on the standard template) show

which conclusions depend on wording (Figure 4). $D_{\text{superseded}}$ was positive in all ten split-by-task estimates (3.133 to 6.599 nats), and Δ_{semantic} in nine of ten. Δ_{mention} never favored the superseded assignment. On the standard template, however, it ranged from -0.806 to $+0.163$, with six of eight intervals including zero. The size of the mention advantage depends on wording.

Gate-rule estimates. The replication amendment defined competent cells by each model’s frozen gate. Table 3 uses the cells competent on the confirmation data, the rule used throughout the text. For Qwen3-8B, the gate rule instead selects routing int.+seq. ($n = 256$): $D_{\text{superseded}} = +3.248$, $\Delta_{\text{semantic}} = +1.330$, $\Delta_{\text{mention}} = -1.702$, net switch rate $+0.195$; access int. ($n = 128$): $D_{\text{superseded}} = +4.221$, $\Delta_{\text{semantic}} = -0.132$, $\Delta_{\text{mention}} = -0.740$, net switch rate $+0.273$. For Qwen3-14B, the gate rule selects the same cells.

	Overall	Access	Routing
Failures	892/6144	454/3072	438/3072
Log loss, prevalence	0.414	0.419	0.410
Log loss, pre-decision	0.346	0.323	0.358
AUROC, pre-decision	0.780	0.834	0.729
AUROC, $+R_{\text{easy}}$	0.779	0.833	0.730
Gain from R_{easy}	$+0.0004$	-0.0006	$+0.0013$

Table 6: H3: out-of-fold prediction of strict downstream failure on historical-construction items (history-grouped five-fold cross-validation). 95% intervals for the gain: overall $[-0.0011, 0.0019]$, access $[-0.0020, 0.0008]$, routing $[-0.0022, 0.0047]$. The decision’s own action confidence, a contemporaneous reference that is not available before the decision, reaches AUROC 1.000.

C Reproducibility

Models. Qwen3-8B, Hugging Face revision b968826d9c46dd6066d109eabc6255188de91218; Qwen3-14B, revision 40c069824f4251a91eefaf281ebe4c544efd3e18; Gemma 3 4B (instruction-tuned), revision 093f9f388b31de276ce2de164bdc2081324b9767. All ran in float32 without quantization, with SDPA attention, on one 80 GB H100 GPU, using Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019). Licenses: Qwen3 (Apache 2.0) and Gemma 3 (Gemma Terms of Use), used for non-commercial research within their terms.

Artifacts. Each dataset is regenerated deterministically from its seed and checked against

Model	Cell	Direct	No hist.	History	Comp.
Qwen3-8B	routing seq.	0.977	0.938	0.779	yes
	routing int.	0.986	0.938	0.779	yes
	routing comp.	0.945	0.938	0.752	no
	access seq.	0.889	0.906	0.822	no
	access int.	0.953	0.938	0.877	yes
	access comp.	0.953	0.859	0.883	no
Qwen3-14B	routing seq.	0.996	0.922	0.854	yes
	routing int.	1.000	1.000	0.846	yes
	routing comp.	0.997	1.000	0.814	yes
	access seq.	0.991	0.781	0.729	no
	access int.	0.977	0.828	0.748	no
	access comp.	0.946	0.828	0.697	no
Gemma 3 4B	routing seq.	0.939	0.688	0.645	no
	routing int.	0.964	0.609	0.488	no
	routing comp.	0.916	0.625	0.533	no
	access seq.	0.945	0.719	0.678	no
	access int.	0.936	0.609	0.676	no
	access comp.	0.946	0.672	0.641	no

Table 7: Frozen gate split (384 histories, identical rows for every model): strict direct current-state retrieval, current-only decision accuracy, decision accuracy with history, and whether the cell is competent (direct ≥ 0.95 , no history ≥ 0.90 , invalid ≤ 0.05). This gate decided eligibility to generate confirmation data. Labels computed on the confirmation data (Table 1, Table 3) can differ: Qwen3-8B access int. is competent here and not competent on the confirmation data; Qwen3-8B routing comp. is not competent here and competent on the confirmation data.

its manifest. Each score file carries per-record checksums and a completion record binding the scores, retrieval scores, run manifest, tokenizer audit, code hash and configuration hash. Analyses refuse incomplete or altered runs. Confirmation: seed 82003, dataset hash 3e972257. . . , code hash c6648aba. . . , configuration hash 78d09215. . . , score file SHA-256 b159b727. . . . The protocol was frozen on 2026-10-08 (UTC), before the gate and confirmation splits were generated. Every number and figure in this paper is produced from the sealed analysis reports by one script, which records the hash of each input.

Compute. Qwen3-8B’s confirmation scoring took 6.27 GPU hours (2.67 s per record, including two retrieval queries, one historical query where applicable, two scored action events and greedy generation). Qwen3-14B took 1.12 (development), 2.23 (gate) and 4.48 GPU hours (confirmation), 1.91 s per confirmation record. Both runs used the same GPU type, software versions (PyTorch 2.14.0, Transformers 5.17.0), scorer and prompts (mean

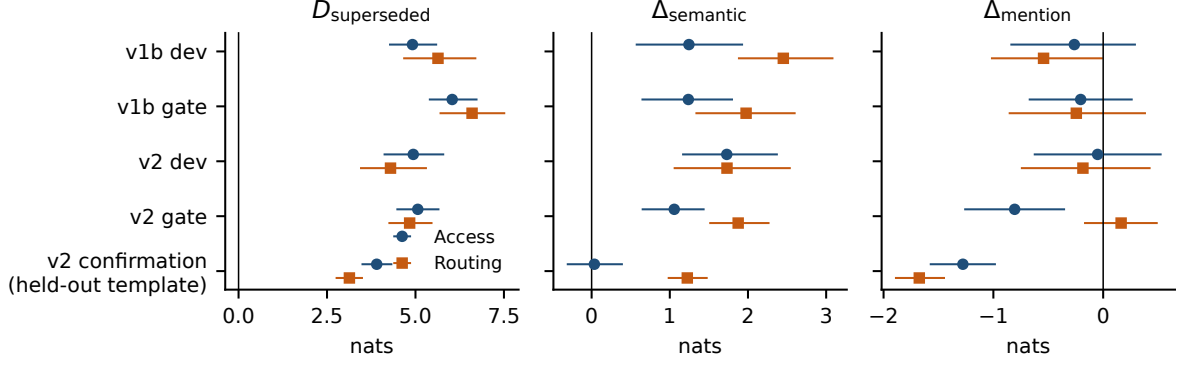


Figure 4: Primary and mention contrasts across all five splits run under the v1b and v2 protocols (95% intervals for development and gate splits, 97.5% for the confirmation’s primary contrasts and 95% for its Δ_{mention}). Development and gate splits used the standard template, and v1b also kept the Answer : line. Only the confirmation is confirmatory.

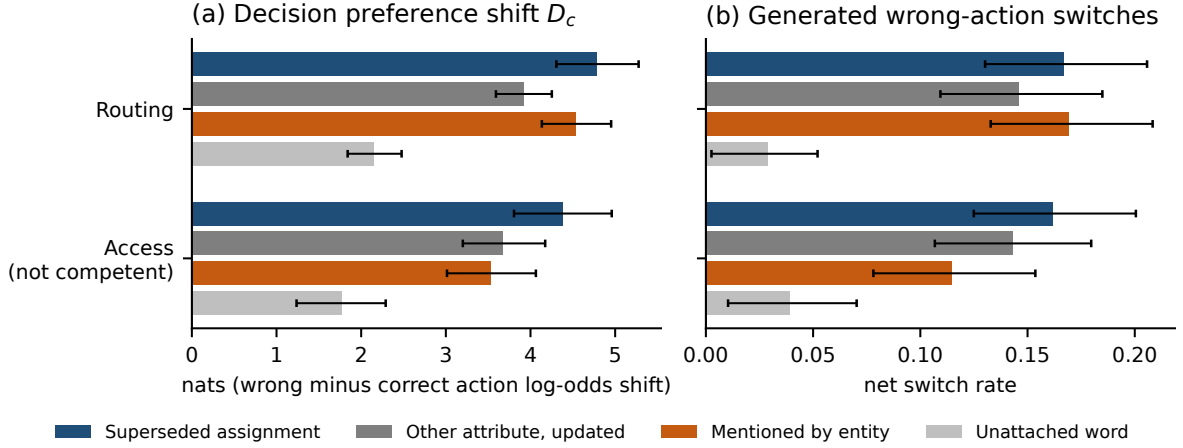


Figure 5: Qwen3-14B: decision effects by construction (confirmation, 384 histories per task, 95% history-bootstrap intervals), as in Figure 2. Routing cells meet the competence criteria on the confirmation data; access cells do not.

199 prompt tokens). The times are wall-clock as logged. We did not record GPU sharing or clock state, so we cannot attribute the faster per-record time of the larger model. The whole program, including failed runs, development and gate splits and the ineligible model, used about 32 GPU hours.

Code. The anonymized archive submitted as supplementary software contains the generators, scoring and analysis code, tests, configuration files, the dated protocol records, the sealed analysis reports with their run records, the confirmation datasets and per-record scores of both confirmed models, and the scripts that derive every number, table and figure in this paper; run from the archive, they reproduce the paper’s derived values exactly.

Model	Construction	Kept	Net switch rate	Switched
Qwen3-8B	superseded	348/384	+0.158 [+0.118, +0.198]	58 / 3
	other attr.	357/384	+0.048 [+0.014, +0.084]	29 / 12
	mention	349/384	+0.278 [+0.232, +0.327]	97 / 0
	unattached	351/384	+0.034 [+0.009, +0.060]	17 / 5
Qwen3-14B	superseded	376/384	+0.165 [+0.128, +0.205]	65 / 3
	other attr.	367/384	+0.136 [+0.101, +0.174]	52 / 2
	mention	363/384	+0.168 [+0.132, +0.207]	61 / 0
	unattached	375/384	+0.024 [+0.000, +0.051]	16 / 7

Table 8: Routing histories the model demonstrably tracks (criteria of Table 2), per model and construction. Condition (ii), initial-state retrieval, is asked only on the superseded members and is applied at the history level to all four constructions. Net switch rate with 95% history-bootstrap intervals; switched: histories switched to the wrong / to the correct action.